

ERICK ALBERTO SÁNCHEZ ARANDA

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EDUCATION

Tecnológico de Monterrey (ITESM)

B.S. in Computer Science and Technology – GPA: 9.0/10.0 (equivalent to 3.6/4.0)

Aug. 2024 – Expected Jun. 2028

Guadalajara, Jalisco

- Relevant Coursework: Data Structures and Algorithms, Computer Architecture, IoT Software Engineering, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, C++, Dart, TypeScript, JavaScript, SQL

ML & Data: PyTorch, NumPy, Pandas, scikit-learn

Frameworks & Backend: Flutter, Next.js, FastAPI, AWS (Cognito, Lambda, EC2), Firebase, Supabase, PostgreSQL

Infrastructure & Tools: Docker, Git, GitHub Actions, Linux (RHEL/Ubuntu), Cloudflare (Zero Trust, Tunnels), JetBrains IDEs

CERTIFICATIONS

AWS Certified Cloud Practitioner

Aug. 2025

EXPERIENCE

Administrative Assistant

Transportes Sánchez Barcelata

Aug. 2022 – Present

Guadalajara, Jalisco

- Built FDM (Factura Diesel Manager) on own initiative, a Python automation tool using PySide6 and Selenium that downloads and parses multi-page diesel invoices from Mexico's tax authority (SAT). Automated a recurring manual reconciliation workflow previously handled by administrative staff. Packaged as a Windows installer via PyInstaller + NSIS for distribution to non-technical staff.
- Built a Python tool to bulk-download payroll invoices (PDF + XML) from SAT for a full fiscal-year compliance audit on a tight deadline. Adapted FDM's architecture and implemented batch scheduling with retry logic to handle SAT's daily download throttling.

LEADERSHIP & ACTIVITIES

PawnGuard – Cybersecurity Student Group

Infrastructure Staff

Oct. 2025 – Present

Guadalajara, Jalisco

- Engineered CI/CD deployment pipeline (GitHub Actions, Docker) to provision CTFd infrastructure for Hack2Dawn, an internal jeopardy-style CTF competition sponsored by MLH, 330ohms, and Google Cloud.
- Developed two OSINT challenge labs for CTF events: GitHub commit forensics and search-engine reconnaissance exercises. Co-organized event logistics with the rest of the staff team.

PROJECTS

Traffic Sign Recognition Research – CNN Image Classification | PyTorch, Python

Undergraduate Research Assistant – Tec de Monterrey

2026 – Present

- Designed and trained convolutional neural networks on the GTSRB dataset (German Traffic Sign Recognition Benchmark, 43 classes), achieving 93% test accuracy with a baseline CNN (~628K parameters) trained in 10 epochs. Advisor: Prof. Atoany Fierro Radilla.
- Self-hosted GPU training infrastructure on RHEL Linux server (dual RTX 3060, NVIDIA Container Toolkit, Docker), exposed via Cloudflare Tunnel for secure remote access.

Cuentas Claras LATAM – Expense-Splitting Mobile App | Flutter, AWS, Supabase

Solo Developer – cuentasclaraslatam.com

2025 – Present

- Architected cross-platform mobile application (iOS & Android) using Flutter/Dart with DEV/PROD flavors, designed for users ages 13+ across LATAM without requiring bank account integration.
- Integrated AWS Cognito for federated authentication (Google, Apple) with a Pre-Token Generation Lambda injecting custom JWT claims, plus Supabase Edge Functions and Firebase for a multi-cloud serverless backend.
- Implemented database security hardening: row-level security on all tables, 80+ SECURITY DEFINER stored procedures with caller validation, pgcrypto encryption, and rate-limited endpoints.

Zenalyse – IoT Environmental Monitoring System | Raspberry Pi, AWS, Next.js

Skilliket Symposium – 1st Place

Nov. 2025

- Designed IoT monitoring system with Raspberry Pi and sensors (DHT11, MQ-135, PIR, LDR) running Python scripts for 5-second polling, correlating environmental data with user stress levels via an integrated survey.
- Deployed backend on AWS EC2 with PostgreSQL and Next.js (TypeScript, Tailwind CSS), exposed via Cloudflare Tunnel for secure global access.

LANGUAGES

Spanish (native), English (Professional working proficiency, B2 First — Cambridge English)